

Introduction to Cellular Communication Networks – Fall 2025

Project Title

Single Cell Wireless System Simulation

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1. Introduction

Wireless cellular systems must support a growing number of users while maintaining acceptable quality of service. In a real cellular network, users initiate calls randomly, experience varying signal strength due to propagation effects, and compete for limited radio resources. As a result, some calls may be blocked or dropped.

The purpose of this project is to simulate a single-cell wireless communication system and study its performance under realistic radio channel conditions. The simulation focuses on downlink transmission, where the base station (BS) transmits to multiple mobile stations (MSs). The system includes radio propagation effects such as path loss, shadowing, and Rayleigh fading, as well as call arrival, call duration, blocking, and dropping behavior.

The main goals of this project are:

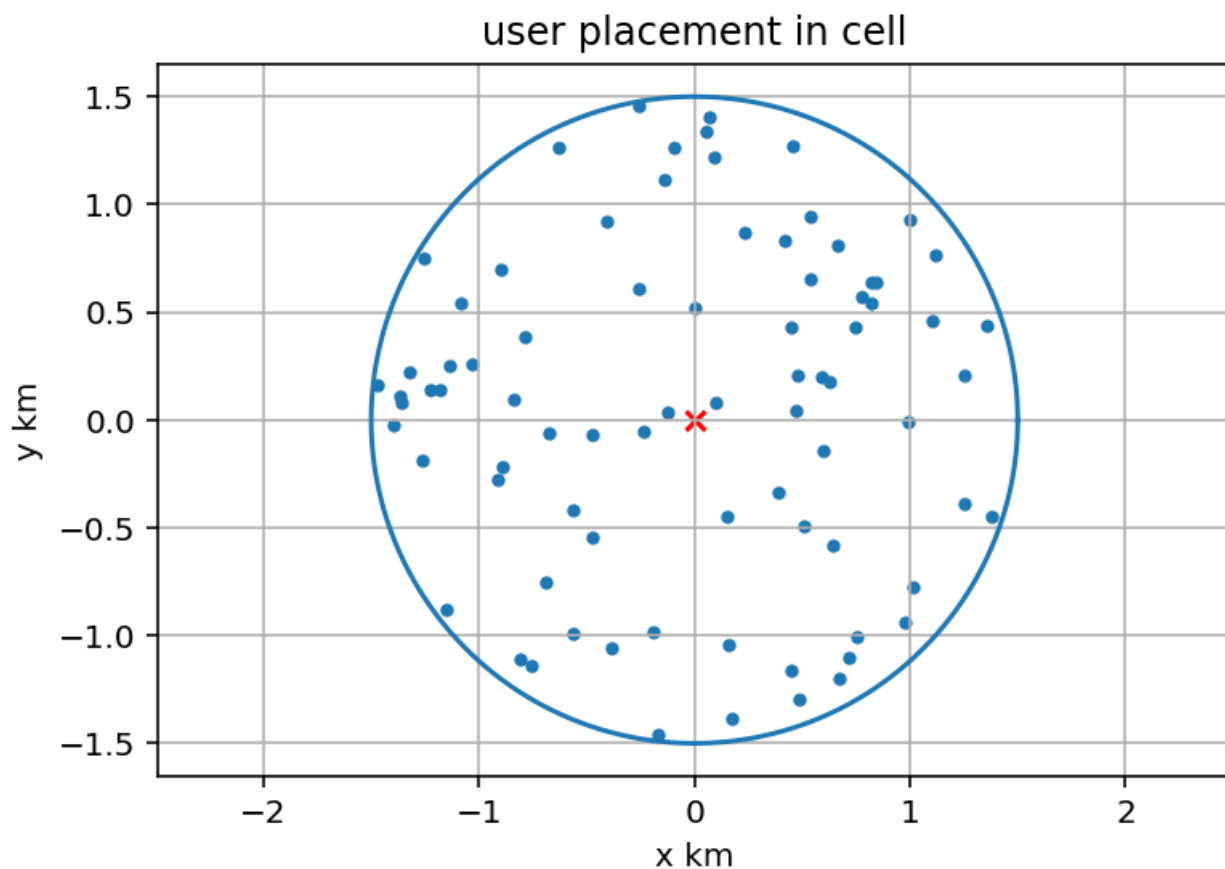
- To model a realistic single-cell system with randomly placed users
- Computing call blocking and call dropping probabilities
- To evaluate the Grade of Service (GOS) of the system
- To determine how many users can be supported at approximately 2% GOS
- To improve system performance by modifying system parameters without increasing the number of channels

2. System Description

2.1 Cell Geometry and User Placement

The simulated system consists of a single circular cell with radius 1.5 km. The base station is located at the center of the cell. Mobile stations are randomly placed inside the cell area. User positions are generated uniformly over the circular area to ensure equal probability of being located anywhere within the cell.

Each mobile station is assigned x- and y-coordinates for visualization purposes, while the distance from the base station is used to compute path loss.



2.2 Base Station Parameters

The base station parameters used in the simulation are:

Parameter	Value
Transmit power	10 W (baseline), 40 W (modified)
Antenna gain	8 dB
Feeder loss	4 dB
Antenna height	30 m
Number of channels	50

2.3 Mobile Station Parameters

All mobile stations are assumed to have identical characteristics:

Parameter	Value
Sensitivity	-102 dBm
Antenna gain	3 dB
Body/orientation loss	3 dB
Antenna height	1.5 m
Call arrival rate	0.017 calls/min
Call duration	3 minutes

3. Wireless Channel Modeling

3.1 Path Loss Modeling

Large-scale path loss is calculated using the Okumura–Hata propagation model, which is suitable for urban cellular environments. The model depends on the carrier frequency, distance between BS and MS, and antenna heights. The path loss is calculated once per user and remains constant during the simulation.

3.2 Shadowing

Shadowing represents slow variations in signal power caused by obstacles such as buildings. It is modeled as a log-normal random variable with a standard deviation of 5 dB. Each mobile station is assigned a fixed shadowing value that does not change over time.

3.3 Rayleigh Fading

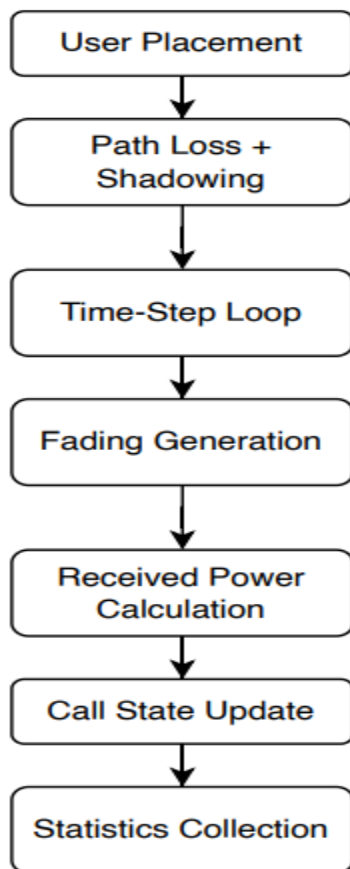
Rayleigh fading models fast signal fluctuations due to multipath propagation. At every simulation time step, a new fading value is generated independently for each mobile station. This effect is responsible for sudden drops in received signal power and is a major cause of call dropping.

4. Simulation Design and Structure

The simulator is designed using a modular structure that follows the assignment instructions closely.

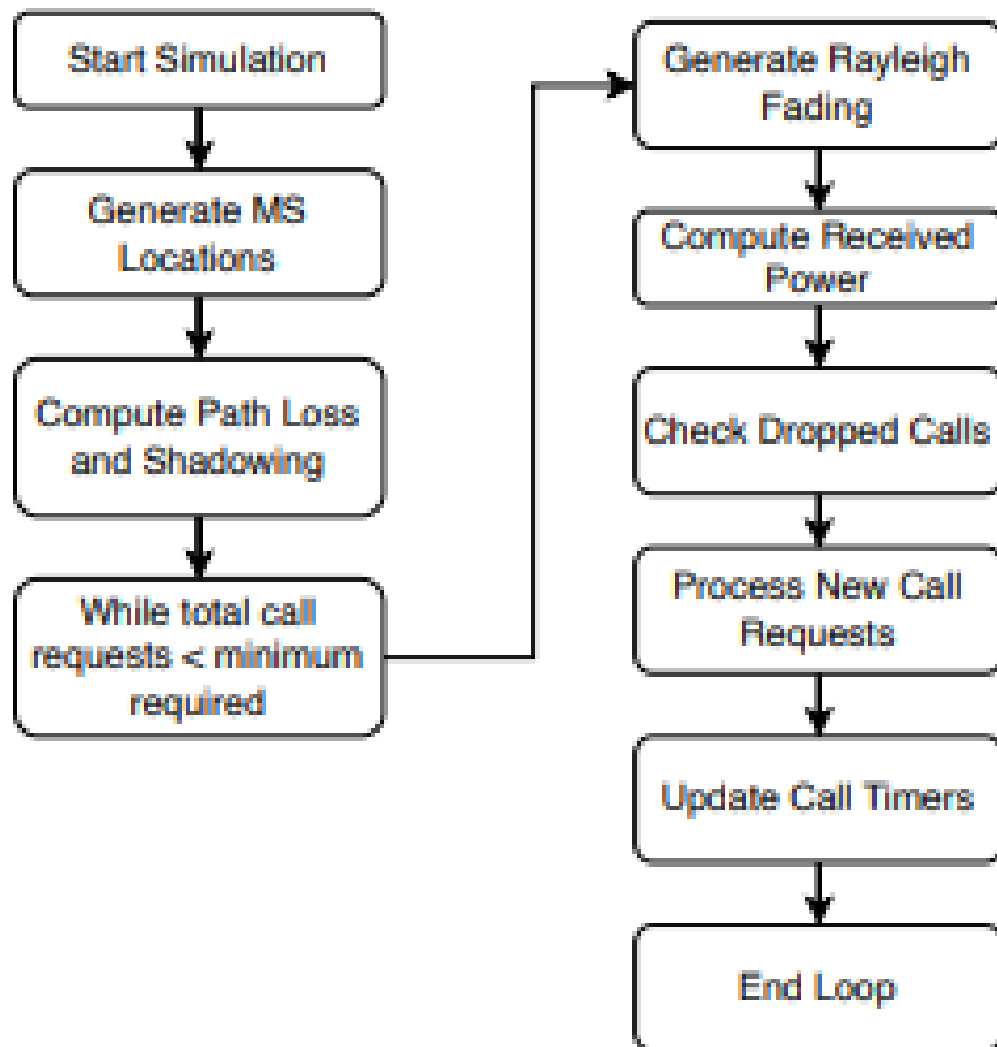
4.1 Block Diagram

The overall structure of the simulator can be represented by the following blocks:



4.2 Simulation Flowchart

The simulation operates in discrete time steps of 1 second. The main flow of the simulation is shown below:

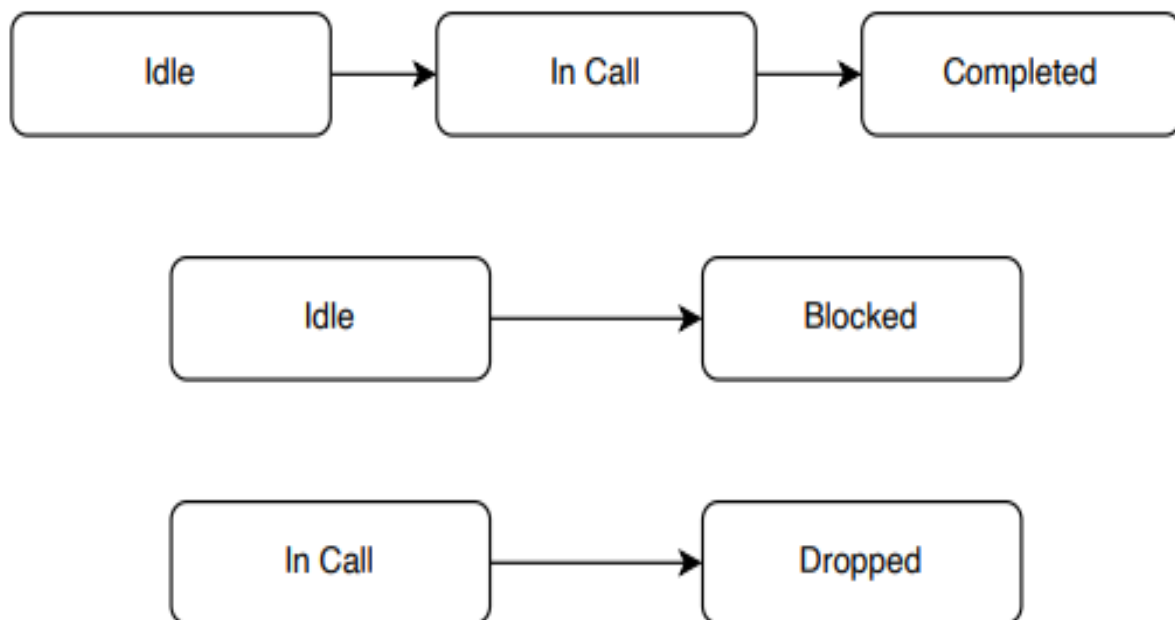


4.3 Call State Transition Diagram

Each mobile station can exist in one of the following states:

- Idle
- In Call
- Dropped
- Completed

State transitions occur as follows:



5. Call Processing Logic

5.1 Call Generation

At each time step, idle users generate a call request with probability:

$$P = \lambda \cdot \Delta t$$

where λ is the call arrival rate and Δt is the time step.

5.2 Call Blocking

A call request is blocked if:

The received power is below the MS sensitivity

No BS channels are available

Blocked calls contribute to the blocking probability.

5.3 Call Dropping

An ongoing call is dropped immediately if the received power falls below the sensitivity threshold due to fading.

5.4 Call Completion

If a call reaches its full duration, it ends successfully and releases its channel. This event is not counted as blocked or dropped.

6. Simulation Experiments

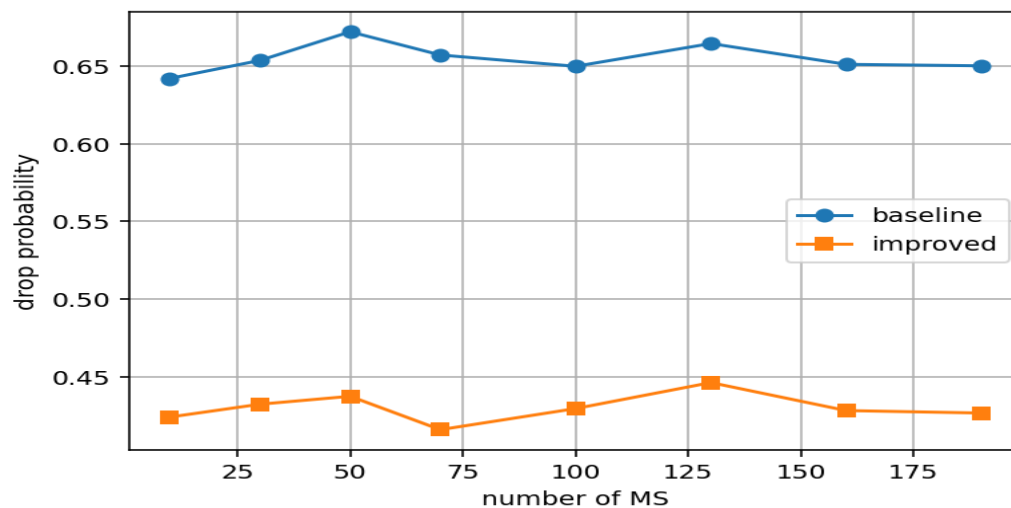
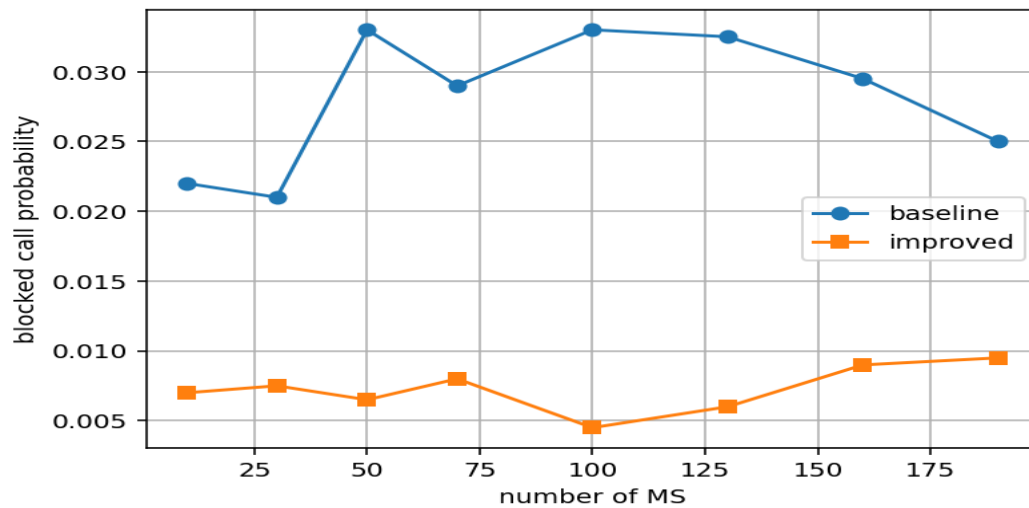
6.1 Baseline System

The baseline system uses a BS transmit power of 10 W. Simulations are performed for increasing numbers of mobile stations. Each configuration is repeated multiple times to obtain reliable averages.

6.2 Modified-Parameter System

To improve system performance without increasing the number of channels, the BS transmit power is increased to 40 W. All other parameters remain unchanged.

7. Simulation Results



Observations:

Blocking probability increases as user load increases

Dropping is mainly caused by fading

Increasing BS transmit power reduces both blocking and dropping

Due to the randomness of fading and user placement, small variations between simulation runs are expected.

8. Grade of Service (GOS) Analysis

Grade of Service is defined as the probability of blocked calls. The target GOS is 2%.

The simulator includes an automatic search that identifies the number of mobile stations that results in a blocking probability closest to 2%.

Based on the simulation results, the system can support approximately NMS = 10 mobile stations at a Grade of Service close to 2%. For higher numbers of users, the blocking probability exceeds the 2% target.

9. System Improvement Discussion

The baseline results show that call blocking and call dropping increase as the number of mobile stations increases. This is mainly due to limited channel availability and low received signal power, especially for users farther from the base station.

To improve the Grade of Service without increasing the number of channels, the base station transmit power was increased from 10 W to 40 W. This increase improves the received signal strength at the mobile stations, which reduces both call blocking due to insufficient power and call dropping caused by fading.

The simulation results confirm that higher transmit power allows the system to support more users while maintaining a similar Grade of Service. Other possible improvements include increasing antenna height or gain, reducing feeder loss, or improving receiver sensitivity.

10. Conclusion

In this project, a single-cell wireless communication system was simulated to study call blocking and call dropping behavior under realistic radio channel conditions. The simulation included path loss using the Okumura-Hata model, log-normal shadowing, and Rayleigh fading, as well as call arrival and call duration modeling.

The results show that both blocking and dropping probabilities depend strongly on the number of mobile stations in the cell and the received signal strength. As the user load increases, limited channel availability and fading effects lead to higher blocking and dropping rates. The Grade of Service analysis indicates that the system can support approximately $NMS = 10$ users at a blocking probability close to 2% under the baseline configuration.

System performance was improved by increasing the base station transmit power without adding new channels. This modification reduced both call blocking and call dropping, allowing more users to be supported at the same Grade of Service level. Overall, the simulation demonstrates how system parameters and radio propagation effects influence cellular network performance and highlights the importance of careful system design.

11. References

1. Hata, M., "Empirical Formula for Propagation Loss in Land Mobile Radio Services," IEEE Transactions
2. Okumura, Y., "Field Strength and its Variability in VHF and UHF Land-Mobile Radio Service"